

Analytics Unit, Integration, and System Testing

Retested September 26, 2026 | Controlled fixture reference date: September 1, 2026

AUTOMATED RETEST: PASS

The earlier summary omitted **Donor Availability Insights**. That was a genuine test-coverage gap: four donor records existed in the controlled fixture, but their predictive insight and prescriptive recommendation were not asserted. The calculation is now a shared analytics function used by both the Reports page and the validation suite.

Scope corrected. This retest now covers every predictive and prescriptive feature displayed on Admin > Reports & Analytics: shortage forecasts, usage trends, donor availability insights, donor campaign recommendations, and inventory transfer recommendations.

UNIT LOGIC

19/19

PASS

CONTROLLED FLOW

128/128

PASS

API INTEGRATION

6/6

PASS

Test-level results

Level	Result	Evidence
Unit	PASS	19 analytics logic regression checks, including donor recovery rules and forecast-window boundaries.
Integration	PASS	128 controlled analytics checks plus 6 API integration checks.
System smoke	PASS	Frontend lint and production build passed. The served /admin/reports route returned HTTP 200 and delivered the donor report module.
Visual browser	NOT EXECUTED	The in-app browser runtime was unavailable. This report does not claim a visual-layout or click-interaction pass.

Analytics coverage

Type	Feature	What was checked
Predictive	Blood shortage forecast	Stock, expiry, usage, demand, days remaining, shortage status.
Predictive	Blood usage trends	Prior/current usage, change, seven-day demand, trend and risk labels.
Predictive	Donor availability	Eligibility now, 7/30-day recovery, percentage, peak month, availability class.
Prescriptive	Donor outreach	Campaign/reminder recommendation derived from the availability result.
Prescriptive	Inventory transfers	Priority, destination stock, source, amount, reserve, recommendation order.

ADDED COVERAGE

Donor Availability Insights

Exact expected and actual output for the controlled 30-day case

ELIGIBLE NOW

2

of 4 donors

AVAILABLE IN 30 DAYS

3

75% of donors

CLASSIFICATION

High

Availability

Output	Expected and actual
Registered donors	4
Eligible now	2
Newly eligible within 30 days	1
Newly eligible within 7 days	0
Can donate within 30 days	3 donors (75%)
Peak upcoming eligibility month	September (1 donor)
Availability label	High Availability

Validated recommendation: Schedule blood donation drives in September to maximize participation when the largest group finishes recovery.

Donor-specific unit cases

- Whole-blood 90-day, platelet 14-day, and plasma 28-day recovery intervals.
- Donors with missing or invalid last-donation dates.
- Different results for 7-day and 30-day forecast windows.
- Empty donor data returns a Low Availability label and an actionable registration recommendation.

Consistency evidence

- Output correctness: **120/120 passed.**
- Five exact repeated-run comparisons passed.
- An equivalent cloned input produced an identical result.
- Input records remained unchanged after processing.
- Complete output SHA-256:
728d5cca06d8f198340417d77d726015c960fcd6ccb6b8c1aa3563f88e91c73d

Interpretation

A pass confirms that the tested code transforms the controlled inputs into the recorded expected outputs consistently. It does not establish clinical effectiveness or a real-world predictive-accuracy percentage. Visual rendering remains unverified until a browser runtime is available.

Detailed artifacts: backend/simulation/output/validation-report.json and docs/Analytics_Functional_Consistency_Testing.md.